MFT-GAN: A Multiscale Feature-Guided Transformer Network for Unsupervised Hyperspectral Pansharpening

Yanli Shang, Jianjun Liu, *Member, IEEE*, Jingyi Zhang, and Zebin Wu, *Senior Member, IEEE*

***Abstract*— Unsupervised learning, learning data distributions without needing labeled samples, is a particularly promising approach for solving the challenging task of hyperspectral pansharpening. Inspired by the above, we introduce an innovative generative adversarial network framework (named MFT-GAN) that incorporates a transformer network and multiscale interaction technology. Specifically, MFT-GAN is constructed of one generator and two discriminators. The generator is composed of a multiscale feature guidance branch (MFGB) and a feature interaction fusion branch (FIFB). The former aims at extracting spectral and spatial information branches from low-resolution hyperspectral (LRHS) and panchromatic (PAN) images by convolutional sampling operations, respectively. The latter accomplishes the interaction of spectral and spatial information at different scales through multiscale interaction technology. Then, the multiscale interaction results are fused by level-by-level convolutional sampling and pixel summation operations to generate high-resolution hyperspectral (HRHS) images. The discriminators are composed of a spectral transformer discriminator and a spatial transformer discriminator, designed to maintain structural and parametric balance and learn long-range spectral and spatial correlation relationships. In addition, hybrid loss functions are used to complete the adversarial training of the MFT-GAN to improve its performance. The experiments on simulated and real datasets further demonstrated the effectiveness of the proposed MFT-GAN method.**

***Index Terms*— Feature interaction fusion, generative adversarial network, multiscale feature guidance, spectral and spatial transformer, unsupervised hyperspectral pansharpening.**

I. Introduction

HYPERSPECTRAL pansharpening [1] is a technique that produces high-resolution hyperspectral (HRHS) images by integrating low-resolution hyperspectral (LRHS) images with panchromatic (PAN) images. Remote sensing imaging systems can provide only high spatial resolution images (e.g., PAN images) or high spectral resolution images (e.g., LRHS images), driving the development of sharpening techniques. Moreover, the information contained in images can be improved by combining hyperspectral information from LRHS images with high-resolution PAN information from PAN images, which helps analyze and identify target features more accurately. Pansharpening has contributed to many fields, such as classification [2], [3], [4], unmixing [5], [6], [7], [8], and detection [9], [10], [11].

Currently, methods for image pansharpening can generally be categorized into two classes: traditional methods and deep-learning-based methods. Component substitution (CS), multiresolution analysis (MRA), and model-based methods are the classic examples of traditional methods. The CS-based pansharpening methods can retain spatial detail information and are easily implemented [12], [13], [14]. MRA-based pansharpening methods can avoid the problem of spectral distortion via detailed injection techniques [15], [16]. The model-based methods construct a mathematical model for pansharpening and solve it to obtain the final fused result [17], [18], [19], [20]. Although the CS and MRA methods enable the final results to retain most of the spatial information, they often lead to more severe spectral distortions. Model-based methods can simultaneously retain spatial and spectral information, but the solution process is complicated or even difficult to solve.

Deep learning is dedicated to solving a variety of complex tasks and problems by simulating the neural networks of the human brain. At present, the development of deep learning has led to many significant reforms in the field of image processing. It has promising applications in various fields, including image classification [21], [22], [23], image segmentation [24], and image superresolution [25], [26], [27], [28]. Currently, deep-learning-based pansharpening methods are divided into supervised and unsupervised methods. The supervised learning methods train an end-to-end network by making full use of labeled data to obtain better quality fused images. Unsupervised learning methods generally do not require labeled data for the training data. Pansharpening relies on the intrinsic structure and statistical information of the input images. However, these methods face several challenges.

1) *Challenges of Supervised Learning Methods:* When the ground truth is not available, supervised learning

Manuscript received 5 January 2024; revised 30 April 2024; accepted 11 May 2024. Date of publication 16 May 2024; date of current version 28 May 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62071204, in part by the Natural Science Foundation of Jiangsu Province under Grant BK20201338, and in part by the Fundamental Research Funds for Central Universities under Grant JUSRP221026. *(Corresponding authors: Jianjun Liu; Zebin Wu.)*

Yanli Shang, Jianjun Liu, and Jingyi Zhang are with Jiangsu Provincial Engineering Laboratory for Pattern Recognition and Computational Intelligence, Jiangnan University, Wuxi 214122, China (e-mail: aaysyl@126.com; liuofficial@163.com; 18438859733@163.com).

Zebin Wu is with the School of Computer Science, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: zebin.wu@gmail.com).

Digital Object Identifier 10.1109/TGRS.2024.3402058

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:40:21 UTC from IEEE Xplore. Restrictions apply.

methods complete model training by downsampling the training data resolution [29], [30]. There is likely to be a distribution mismatch between the training data and the test data, and this limitation significantly reduces the overall performance of the model.

2) *Limitations of Existing Unsupervised Learning Methods:* In the existing generative adversarial network (GAN)-based unsupervised pansharpening methods, the long-range interaction fusion between image features (learning data distribution) is not considered in the generator. The discriminator is constructed based on universal classifiers instead of matching the structure of the generator to be designed for accomplishing discriminative and adversarial training tasks.

Inspired by the above, this article attempts to construct an unsupervised transformer framework based on the foundation of GANs. In detail, we propose a multiscale feature-guided transformer framework based on a GAN for unsupervised hyperspectral pansharpening (MFT-GAN). This approach takes advantage of transformer and multiscale interaction technology to realize interactions between features on the same scale and fusions between features on different scales. In particular, the generator is a multiscale feature-guided transformer network that consists of a multiscale feature guidance branch (MFGB) and a feature interaction fusion branch (FIFB). The MFGB can extract multiscale spectral information branches from LRHS images and multiscale spatial information branches from PAN images via convolutional neural networks (CNNs). In FIFB, we design multihead cross-attention modules for interactively fusing the spectral and spatial information of the MFGB to generate HRHS images.

In the GAN framework, factors such as the structural design and tradeoffs of the generator and discriminator and the challenges of configuring the training process may lead to instability in GAN training. To guarantee the structural balance and parameter balance of the framework in continuous adversarial training, we design a spectral transformer discriminator and a spatial transformer discriminator. These methods acquire global spectral and spatial information from an image, improving the spectral and spatial similarity between the fused image and two observed images. Finally, we perform adversarial learning via MFT-GAN with a hybrid loss function. The summarization of our contribution is as follows.

1) We design a new framework called MFT-GAN (which consists of a generator and two discriminators), which organically combines the GAN framework and transformer networks to improve unsupervised hyperspectral pansharpening performance.
2) In the generator, we design two branches, the MFGB and the FIFB, for multiscale feature interaction and fusion. The MFGB uses CNNs to obtain multiscale information from LRHS and PAN images. The FIFB is embedded with several multihead cross-attention modules that can interact and fuse information at different scales.
3) To confront the discriminator with the generator and reach a certain structural and parametric balance, we design spectral and spatial transformer discriminators to make the generated HRHS images spectrally and spatially closer to the original images. In addition, continuous adversarial training is accomplished by hybrid loss functions to improve performance.

II. Related Works

At present, hyperspectral pansharpening methods can be categorized into traditional and deep-learning-based methods. The deep-learning-based methods include supervised and unsupervised methods.

A. Traditional Approaches

The traditional pansharpening methods are organized into three categories: CS, MRA, and model-based methods. The CS methods usually decompose the LRHS image into spectral and spatial components and then replace the spatial components with the PAN image, which is later inverted to the original space by inversion. The commonly used CS algorithms include Gram–Schmidt (GS) [31], GS-adaptive (GSA) [14], and principal component analysis [12]. The MRA method extracts spatial details from the PAN image via a multiscale transform before injecting them into the upsampled low-resolution multispectral (LRMS) image. These methods include Laplace pyramids [32], curvelet transforms [33], high-pass filtering [34] and additive wavelet brightness ratios [34]. Model-based methods are based on optimization theory, which regards image fusion as an ill-posed optimization problem; solving this problem usually requires optimization algorithms [18], [35], [36], [37]. For example, the P + XS [18] method fuses LRMS and PAN images by linearly downscaling high-resolution multispectral (HRMS) images along a linear degradation of spatial and spectral dimensions. Yokoya et al. [36] introduced the coupled nonnegative matrix factorization (CNMF), which can form HRHS images with the advantages of both the datasets by combining the endmember matrix of LRHS images and the abundance matrix of HRMS images. To avoid the complex step of iteratively solving the optimization algorithm, Wei et al. [38] proposed a fast fusion of multi-band images based on solving a sylvester squation (FUSE). In this approach, the corresponding Sylvester equation is constructed using a circulant matrix and a downsampling matrix, allowing the fusion result to be directly obtained by solving the equation. In general, CS approaches can be easily implemented but suffer from spectral loss in fused images. MRA approaches can avoid the problem of spectral loss but often lose high-resolution information. Model-based approaches can produce high-quality fused images, but model-based methods usually involve complex computational processes that may be difficult to solve.

B. Deep-Learning-Based Approaches

1) Supervised Approaches: In supervised methods, the goal is to construct a network that makes inferences from the input data and predicts the corresponding output labels. CNN-based deep networks are currently more mature than other methods and have been successfully applied to pansharpening. Later, researchers developed many techniques to improve CNN

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:40:21 UTC from IEEE Xplore. Restrictions apply.

performance further, including residual techniques, dense networks, detail injection, and attention mechanisms. Although these techniques have yielded excellent results, physical interpretations of pansharpening are lacking. To address this issue, several superior methods have been proposed, including deep unfolding technique [39], [40], [41], transformer [42], [43], [44], [45], [46], multiscale feature extraction and fusion technique [47], [48], [49], and diffusion model [50], [51].

The deep unfolding technique aims to compensate for the lack of interpretability in pure networks. For instance, Xie et al. [39] developed an HRMS/LRHS fusion model by integrating a generalized model of low-resolution images with the low-rank prior knowledge of HRHS images. The fusion model was solved by unfolding an approximate gradient algorithm to construct a deep network. Xu et al. [40] presented two suboptimization problems involving pansharpening and simulated the solutions of the two suboptimization problems using two network blocks alternately stacked with deep networks.

The transformer is a deep network built on self-attention mechanisms, capable of extracting global features from images. For instance, Meng et al. [42] suggested an improved transformer-based pansharpening method, which captures long-range dependencies between stacked MS and PAN images by dividing them into multiple patches for input into a transformer network. To increase the power of the transformer in the visual domain, a hierarchical transformer using shift windows, i.e., the Swin transformer [45], has been developed. Long et al. [46] proposed a superresolution network with a dual self-attention Swin transformer, which can learn spectral sequence information in the neighboring bands of hyperspectral images, where feature correlations are computed using shifted windows in the global and local feature spaces.

Multiscale feature extraction and fusion technique has been at the forefront. It is very flexible and can be combined with both CNNs and transformers. By decomposing images into various features ranging from finer details to larger overall features, it further extracts and integrates them to enhance modeling capabilities. For example, Meng et al. [48] proposed a bandwidth-limited multiscale high-frequency generator that generates high-frequency signals from PAN images across different bandwidths and gradually integrates them into the MS, achieving high-quality restoration of MS images. Jia et al. [49] presented a multiscale spatial–spectral network to extract spectral and spatial features from LRHS and HRMS images, respectively, and fused them to improve the performance and generalization ability of the network.

The diffusion model is an emerging generative model that consists of two main processes: forward noise addition and backward denoising. In the forward process, the random Gaussian noise introduced at each moment is specified as the actual noise images. In the backward denoising process, a mapping relationship between the predicted noise and the actual noise is established, according to which a deep model is designed and trained to recover the noise image and improve the image quality. Currently, diffusion models have been applied in pansharpening tasks. For instance, Meng et al. [50] presented the pansharpening technique PanDiff using the denoising diffusion probabilistic model. The method decomposes the fusion process of PAN and LRMS images into a multistep Markov process and uses U-Net to reconstruct each step of the process from random Gaussian noise. Cao et al. [51] introduced a diffusion model approach with two conditional modulation modules, which mainly consists of coarse-grained style modulation and fine-grained wavelet modulation, by decoupling the coarse-grained style information from the fine-grained frequency information to generate competitive fused images.

2) Unsupervised Approaches: In unsupervised learning, the designed networks mainly learn the hidden structures, patterns, or relationships from data. The unsupervised pansharpening methods include autocoding-based [52], [53], [54], [55], [56], GAN-based [57], [58], [59], [60], model-based [61], [62], [63], [64], and diffusion model [65] methods. For example, an improved sparse denoising self-encoder algorithm for learning the relationship between PAN and LRMS images was introduced by Huang et al. [52]. Uezato et al. [53] presented a guided depth decoder (GDD) network that is used as a general prior, and it uses the multiscale features of the guided images and the depth decoder network to generate the fused images. To fully extract image features, Qu et al. [54] proposed a deep network based on a self-attention mechanism that uses the spectral properties of LRMS images to estimate the spatial variations in the detailed extraction and injection functions to generate HRMS images. Taking advantage of GANs, Ma et al. [57] presented an unsupervised sharpening framework based on a GAN (Pan-GAN) and its generator and discriminator that establishes an adversarial game for retaining abundant spectral and spatial information. Zhou et al. [59] developed a GAN network based on an unsupervised cyclic consistency GAN (UCGAN). In addition, Zheng et al. [61] and Yao et al. [62] proposed using the CNN framework based on the spectral unmixing model to complete adequate unsupervised feature learning. Rui et al. [65] presented a low-rank diffusion model (PLRDiff) that exploits the power of pretrained deep diffusion models and the better generalization of Bayesian methods.

III. Proposed Method

This article focuses on feature interaction and fusion between different scales. The MFT-GAN consists of one generator and two discriminators. The generator contains two branches, MFGB and FIFB, which are designed to perform the tasks of multiscale feature extraction, interaction and fusion. Two transformer-based discriminators have been designed to preserve the spectral and spatial information of two observed images by means of adversarial learning. Fig. 1 shows the MFT-GAN generator and discriminators. For convenience, $\mathbf{Y} \in \mathbb{R}^{B\times h\times w}$ denotes the LRHS image, and $\mathbf{P} \in \mathbb{R}^{1\times H\times W}$ denotes the PAN image. The fused image is represented as $\mathbf{F} \in \mathbb{R}^{B\times H\times W}$ ($H > h$, $W > w$, $H/h = W/w = r$), where $H \times W$ and $h \times w$ are the spatial sizes, B is the number of channels, and r is the ratio of the spatial resolutions of $\mathbf{P}$ and $\mathbf{Y}$. In addition, G denotes the generator, and D_1 and D_2 denote the spectral and spatial discriminators, respectively.

A. Generator Structure

1) Multiscale Feature Guidance Branch: In hyperspectral pansharpening, the input data contain spectral and spatial

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:40:21 UTC from IEEE Xplore. Restrictions apply.

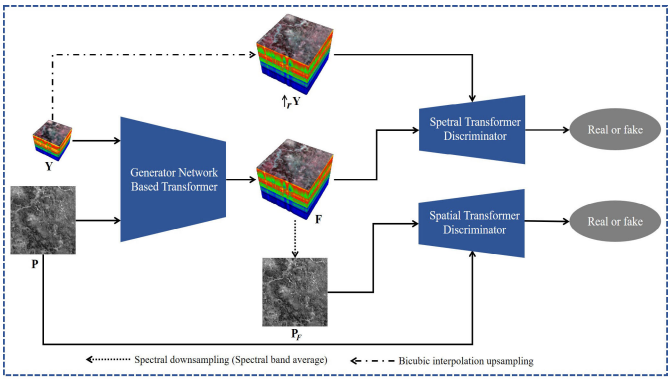


Fig. 1. Overall structure of MFT-GAN.

information (spectral information exists in LRHS images, and spatial information exists in PAN images). To make these two types of information interactively fused over a long range, we first design an MFGB in the generator (see Fig. 2). It is a CNN-based two-branch structure that extracts multiscale spectral and spatial information from LRHS and PAN images, respectively. The two branches with different directions acquire features at different scales by upsampling and downsampling, which are described as follows:

$$\mathbf{Y}_1 = \delta \cdot f_{3\times3}(\mathbf{Y}) \tag{1}$$

$$\mathbf{Y}_i = \delta \cdot f_{3\times3}\left(\mathrm{Up}_i(\mathbf{Y}_{i-1})\right), \quad 1 < i < r(r=4) \tag{2}$$

$$\mathbf{P}_1 = \delta \cdot f_{3\times3}(\mathbf{P}) \tag{3}$$

$$\mathbf{P}_i = \delta \cdot f_{3\times3}(\mathrm{Down}_i(\mathbf{P}_{i-1})), \quad 1 < i < r(r=4) \tag{4}$$

where $f_{3\times3}$ represents a layer of convolution with a kernel size of 3 and a step size of 1, and δ represents the LeakyReLU function. Up represents convolutional upsampling using a layer of 2-D transposed convolution with a kernel size of 2 and a step size of 2. Down represents convolutional downsampling using a layer of 2-D convolution with a kernel size of 3 and a step size of 2. i is used to distinguish images at different scales. As i increases for $\mathbf{Y}$, the image size of $\mathbf{Y}_i$ increases. As i increases for $\mathbf{P}$, the image size of $\mathbf{P}_i$ decreases.

2) Feature Interaction Fusion Branch: After MFGB, multiscale spectral and spatial information can be obtained, which will be used as guiding items driving the implementation of multiscale interaction fusion technology. Specifically, we design a multihead cross-attention module (see Fig. 3), which calculates the long-range correlation between two feature maps at the same scale for the long-range interaction between the information. Using it to compute spectral and spatial information at different scales, we can obtain the interaction results at different scales. Finally, the interaction results at different scales are subjected to the feature aggregation operation to produce the final fusion image (see Fig. 2).

a) Multihead cross-attention module: Inspired by the attention mechanism [66], [67], we develop the multihead cross-attention module to accomplish multiscale interaction. The module has spectral and spatial information inputs ($\mathbf{Y}_i$ and $\mathbf{P}_j$) from the MFGB. First, $\mathbf{Y}_i$ and $\mathbf{P}_j$ are mapped to the same dimension using a layer of convolution with a convolution kernel size of 1 ($f_{1\times1}$), which are reshaped into two sequences. Then, the two sequences are divided into three subsequences of the same dimension: query Q, key K, and value V. The two multiheaded self-attention [66] results are calculated by exchanging their Q (the query Q in one sequence is combined with the key K and value V in the other sequence), which achieves feature interaction. Finally, a feedforward neural network (FNN), residual connection, and pixel addition operations are added to enhance its performance further. In Fig. 2, the multihead cross-attention in the FIFB is expected to compute multiple scales of interaction results between $\mathbf{Y}$ and $\mathbf{P}$ images, which is described and formulated as follows:

$$\mathbf{Y}_i^{\mathrm{Ebedding}} = \mathrm{Reshape}(f_{1\times1}(\mathbf{Y}_i)), \quad i \in [1, 2, \ldots, r-1] \tag{5}$$

$$\mathbf{P}_j^{\mathrm{Ebedding}} = \mathrm{Reshape}\left(f_{1\times1}\left(\mathbf{P}_j\right)\right), \quad j \in [r-1, \ldots, 2, 1] \tag{6}$$

$$Q_1, K_1, V_1 = \mathrm{Chunk}\left(\mathbf{Y}_i^{\mathrm{Ebedding}}\right), \quad i \in [1, 2, \ldots, r-1] \tag{7}$$

$$Q_2, K_2, V_2 = \mathrm{Chunk}\left(\mathbf{P}_j^{\mathrm{Ebedding}}\right), \quad j \in [r-1, \ldots, 2, 1] \tag{8}$$

$$\mathrm{Out}_1 = \mathrm{MultiHead}(Q_2, K_1, V_1) \tag{9}$$

$$\mathrm{Out}_2 = \mathrm{MultiHead}(Q_1, K_2, V_2) \tag{10}$$

$$\mathrm{Temp}_i = \mathrm{Linear}(\mathrm{Out}_1 + \mathrm{Out}_2) + \mathbf{Y}_i^{\mathrm{Ebedding}} + \mathbf{P}_j^{\mathrm{Ebedding}} \tag{11}$$

$$\mathrm{CA}_i = \mathrm{FFN}\left(\mathrm{Temp}_i\right) + \mathrm{Temp}_i \tag{12}$$

where Chunk stands for dividing a sequence into multiple subsequences. MutiHead denotes the multihead self-attention calculation as described in [66]. Temp_i denotes the intermediate output result. In addition, the FNN consists of a two-layer linear connected network and an activation function, LeakyReLU.

b) Feature aggregation: As depicted in Fig. 2, to facilitate the interaction of spectral and spatial information at multiple scales, we use the multihead cross-attention computational strategy. This approach can yield different scales of interaction results. These results are subsequently aggregated to obtain fusion results through level-by-level convolutional upsampling and pixel addition operations

$$\mathrm{FA} = \mathrm{Up}_5\left(\mathrm{Up}_4(\mathrm{CA}_1) + \mathrm{CA}_2\right) + \mathrm{CA}_3 \tag{13}$$

where Up_4 and Up_5 represent convolutional upsampling using a layer of 2-D transposed convolution with a kernel of size 2 and a step of 2.

After the above feature aggregation, the generated image retains the basic spectral and spatial detail information. We use a detailed injection strategy that adds these images to the interpolated upsampled LRHS image to preserve the spectral and spatial information further

$$\mathbf{F} = f_{1\times1}(\mathrm{FA}) + (\uparrow_r \mathbf{Y}) \tag{14}$$

where $f_{1\times1}$ represents a layer of convolution with a kernel size of 1 and a step size of 1. It is used to map FA to the same dimension as $\uparrow_r \mathbf{Y}$.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:40:21 UTC from IEEE Xplore. Restrictions apply.

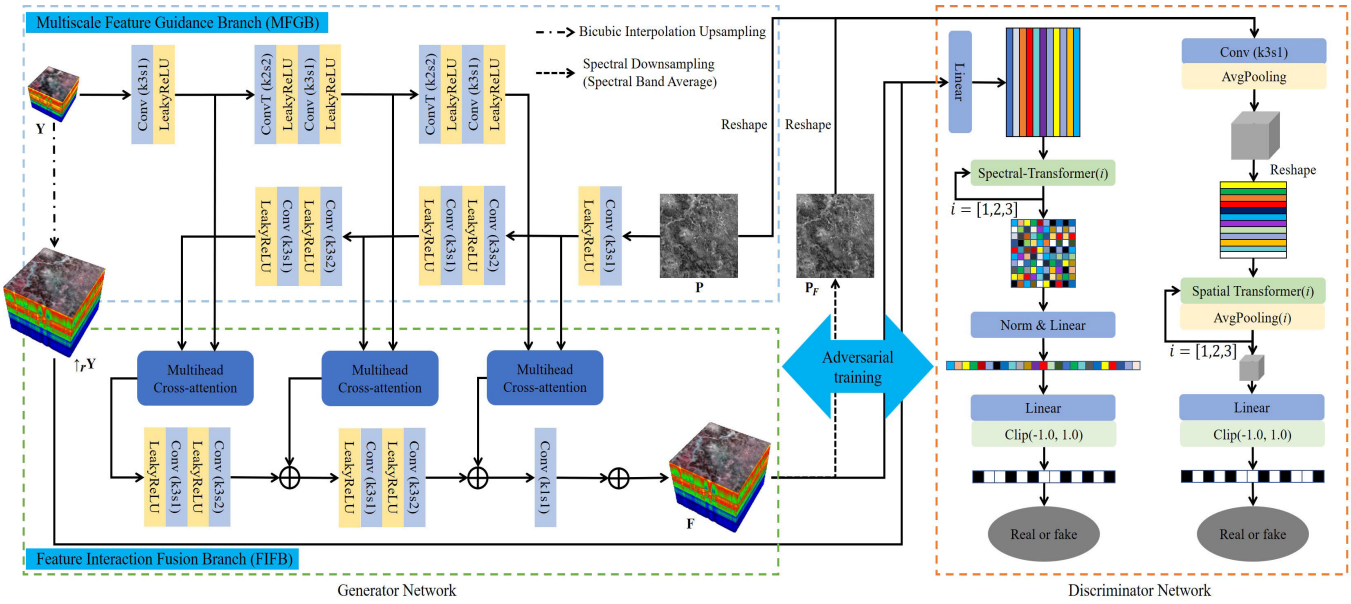


Fig. 2. Detailed structure of the generator and discriminators in MFT-GAN.

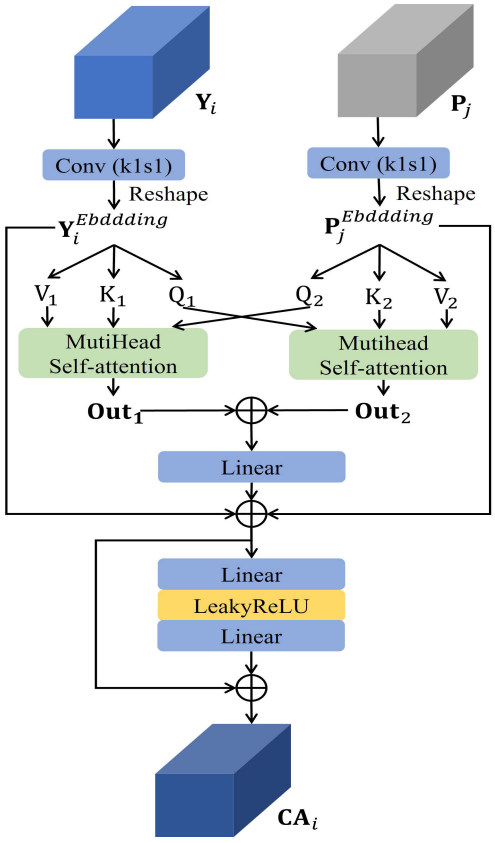


Fig. 3. Illustration of the multihead cross-attention module. ⊕ represents pixel addition.

Procedure 1 Procedure for the Spectral Transformer Discriminator

Input: Generated HRHS image: **F** or interpolated image of LRHS: $\uparrow_r \mathbf{Y}$
Output: Classification result: Ans
1: $\text{Input}_0 = \text{Reshape}(\text{Linear}(\text{Input}))$
2: **for** $i = 1 : r - 1$ **do**
3: $\text{Input}_i = \text{Spectral transformer}_i(\text{Input}_{i-1})$
4: **End for**
5: $\text{Output} = \text{Linear}(\text{Norm}(\text{Input}_{r-1}))$
6: $\text{Ans} = \text{Clip}(\text{Linear}(\text{Output}))$
7: **Return** Ans

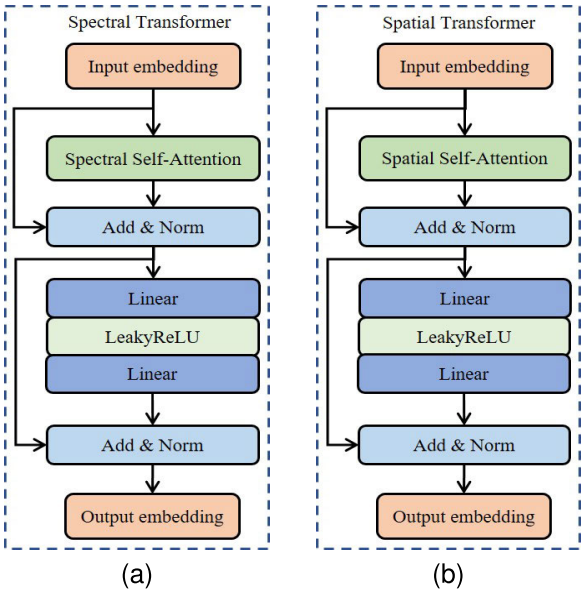


Fig. 4. Structure of spectral and spatial transformers. (a) Spectral transformer. (b) Spatial transformer.

B. Discriminator Structure

For structural and parametric balance, we integrate the transformers into two discriminators (see Fig. 2); the detailed structures of these discriminators are shown in Fig. 4. We can see that spectral and spatial transformers are distinguished using different attention mechanisms, spectral self-attention, and spatial self-attention, which are computed with the same equation [66]. As shown in Fig. 5, the input data have different

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:40:21 UTC from IEEE Xplore. Restrictions apply.

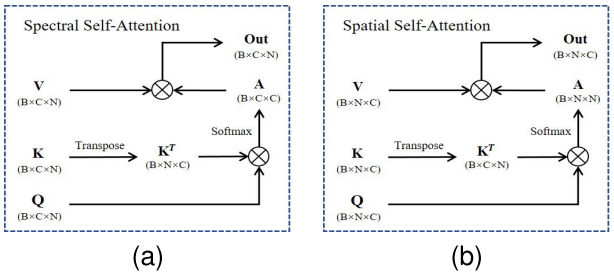


Fig. 5. Description of calculations for spectral and spatial self-attention. (a) Spectral self-attention. (b) Spatial self-attention. C represents the bands of the input data, and N is the spatial size. $\otimes$ is matrix multiplication.

Procedure 2 Procedure for the Spatial Transformer Discriminator

Input: Band average image for generated HRHS: $\mathbf{P}_F$ or PAN image: $\mathbf{P}$
Output: Classification result: Ans
1: $\text{Input}_0 = \text{Reshape}(\text{AvgPooling}(f_{3\times 3}\ (\text{Input})))$
2: **for** $i = 1 : r - 1$ **do**
3: $\quad \text{Input}_i = \text{Spatial transformer}_i(\text{Input}_{i-1})$
4: $\quad \text{Input}_i = \text{AvgPooling}_i(\text{Input}_i)$
5: **End for**
6: $\text{Ans} = \text{Clip}(\text{Linear}(\mathbf{Input}_{r-1}))$
7: Return Ans

dimensions, which calculate different self-attention results, thus allowing long-range spectral features and long-range spatial features to be acquired. Specifically, spectral self-attention calculates the spectral correlation between different pixels, and spatial self-attention calculates the spatial correlation between different bands. This calculation can better preserve the spectral and spatial features.

The spectral transformer discriminator is applied to discriminate the interpolated upsampled LRHS image ($\uparrow_r \mathbf{Y}$) from the generated HRHS image ($\mathbf{F}$) as true or false. In Fig. 2 and Procedure 1, we can see the running process of the spectral transformer discriminator. Unlike the traditional discriminator, the spectral transformer discriminator iteratively calculates the spectral long-range correlation of feature maps through the spectral transformer module. In each iteration of the spectral transformer, the parameters are not shared because the input dimensions are different. This design aims to capture long-range spectral features between different pixel dimensions by gradually decreasing the dimensionality, thus obtaining excellent classification results at the spectral horizon.

The spatial transformer discriminator is used to discriminate between the PAN image ($\mathbf{P}$) and the generated HRHS image ($\mathbf{P}_F$) as true or false. In Fig. 2 and Procedure 2, we can see the running process of the spatial transformer discriminator. Like in the above theory, the spatial transformer discriminator also uses an iterative computational strategy, and the parameters of each iteration are not shared. Long-distance spatial features between different spectra are acquired using a spatial transformer module and an average pooling operation, which achieves excellent classification results at the spatial horizon. Finally, we map the classification results of the two discriminators to a range of -1 to 1 using the clip function,[1] which ensures compatibility with the hinge loss function and contributes to efficient model training.

[1]Compared with other mapping functions, clip function can adjust certain values in the data without destroying the categorical structure of the data.

C. Loss Function

The loss function in MFT-GAN has two components: generator loss (L_G) and discriminator loss (L_{D_1}, L_{D_2}). They are described in detail as follows.

For the generator loss, we build a mixed loss that does not depend on the ground truth for enhancing the generated image quality. The loss includes spectral and spatial reconstruction losses L_{spectral}, L_{spatial}, and adversarial loss L_{adv}. Specifically, the cosine similarity is applied to L_{spectral}, and the mean square error (mse) loss function is used as L_{spatial}. It is formulated as

$$L_G = \lambda_1 L_{\text{spectral}} + \lambda_2 L_{\text{spatial}} + \lambda_3 L_{\text{adv}} \tag{15}$$

$$L_{\text{spectral}} = 1 - \frac{1}{B}\sum_{i=1}^{B} \frac{\mathbf{F}(i,:,:)\cdot \uparrow_r \mathbf{Y}(i,:,:)}{\|\mathbf{F}(i,:,:)\|_2 \|\uparrow_r \mathbf{Y}(i,:,:)\|_2} \tag{16}$$

$$L_{\text{spatial}} = \|\mathbf{P}_F - \mathbf{P}\|_F^2 \tag{17}$$

$$L_{\text{adv}} = \text{HLoss}(D_1(\mathbf{F}), 1) + \text{HLoss}(D_2(\mathbf{P}_F), 1) \tag{18}$$

$$\text{HLoss}(\mathbf{x}, \text{label}) = \begin{cases} \mathbf{x}, & \text{label} = 1 \\ \max(0, 1 - \mathbf{x}), & \text{label} = -1 \end{cases} \tag{19}$$

where $\mathbf{F} = G(\mathbf{Y}, \mathbf{P})$, $\mathbf{P}_F = (1/B)\sum_{i=1}^{B} \mathbf{F}(i,:,:)$ with $\mathbf{F}(i,:,:)$ being the ith band level, and "$\cdot$" represents the dot product. $\mathbf{x}$ represents a 1-D vector, and the symbol "max" represents the vector max calculation. HLoss denotes the hinge loss, which is a commonly applied loss function in GANs [68], [69], [70]. The hinge loss introduces a margin in the computation process, which encourages the discriminator to allow a larger margin in scoring real and fake data, making its classification more certain. Therefore, the losses of spectral and spatial transformer discriminators can be written as

$$L_{D_1} = \text{HLoss}(D_1(\uparrow_r \mathbf{Y}), 1) + \text{HLoss}(D_1(\mathbf{F}), -1) \tag{20}$$

$$L_{D_2} = \text{HLoss}(D_2(\mathbf{P}), 1) + \text{HLoss}(D_2(\mathbf{P}_F), -1). \tag{21}$$

IV. Experimental Results

Our proposed MFT-GAN method is validated through experiments on both the simulated and real data (see Table I). The fused image produced by MFT-GAN is measured by commonly used image quality assessment metrics from different perspectives. Specifically, the evaluation metrics include reference evaluation metrics and nonreference evaluation metrics. The reference evaluation metrics consist of spectral angle mapper (SAM), peak signal-to-noise ratio (PSNR), root mse (RMSE), erreur relative globale adimensionnelle de synthèse (ERGAS), and structural similarity (SSIM) [1], [71]. The no-reference evaluation metrics consist of spatial distortion index (D_S), spectral distortion index (D_lambda), and quality with no reference (QNR) [72].

A. Simulated Datasets

1) CAVE: The CAVE dataset was produced in a laboratory environment by obtaining 32 images of different scenes. For each image, it contains 31 bands with a spectral resolution of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:40:21 UTC from IEEE Xplore. Restrictions apply.

TABLE I
INTRODUCTION OF SIMULATED AND REAL DATASETS AND THE DISTRIBUTION OF TRAINING AND TESTING SAMPLES

Datases	Band	Image Cropping	
		Training	Testing
CAVE	31	PAN: $32 \times 32 \times 1$ LRHS: $8 \times 8 \times 31$ HRHS: $32 \times 32 \times 31$ Patches: 12500	PAN: $512 \times 512 \times 1$ LRHS: $128 \times 128 \times 31$ HRHS: $512 \times 512 \times 31$ Patches: 12
Chicusei	128	PAN: $32 \times 32 \times 1$ LRHS: $8 \times 8 \times 128$ HRHS: $32 \times 32 \times 128$ Patches: 13005	PAN: $256 \times 256 \times 1$ LRHS: $64 \times 64 \times 128$ HRHS: $256 \times 256 \times 128$ Patches: 19
Bostwana	145	PAN: $32 \times 32 \times 1$ LRHS: $8 \times 8 \times 145$ HRHS: $32 \times 32 \times 145$ Patches: 15246	PAN: $128 \times 128 \times 1$ LRHS: $32 \times 32 \times 145$ HRHS: $128 \times 128 \times 145$ Patches: 6
LA	145	PAN: $48 \times 48 \times 1$ LRHS: $12 \times 12 \times 145$ Patches: 11025	PAN: $360 \times 360 \times 1$ LRHS: $90 \times 90 \times 145$ Patches: 1

400–700 nm and a spatial resolution of 512 × 512. In our experiments, the first 20 images are selected for the train set and the last 12 images are selected for the test set.

2) Chikusei: This dataset was captured by the Headwall Hyperspec-VNIR-C imaging sensor in the agricultural and urban areas of Chikushi, Ibaraki, Japan. The hyperspectral dataset contains 128 bands with a spectral range from 363 to 1018 nm. The spatial resolution of this dataset is 2517 × 2335. We crop an image of size 2048 × 2048 along the top left corner and divide it into 64 images of size 128 × 128 by nonoverlapping partitions. Among them, the first 45 images are used for the train set and the last 19 images are used as the test set.

3) Bostwana: The NASA EO-1 satellite acquired the Bostwana dataset from the Okavango Delta in Botswana between 2001 and 2004, and the Hyperion sensor on board EO-1 acquired data in 242 bands within a 7.7-km band at a pixel resolution of 30 m, covering the 400–2500-nm portion of the spectrum within a 10-nm window. To facilitate the experiment, uncalibrated and noisy bands covering water absorption features were removed, leaving 145 bands. The spatial resolution of this dataset is 1496 × 256 and we crop it along the top left corner to create an image of size 1280 × 256. Then, it is divided into 20 images of size 128 × 128, the first 14 images are used as the training set and the last six images are used as the test set.

B. Real Datasets

The Los Angeles (LA) dataset was obtained from Hyperion and the Advanced Land Imager (ALI) sensor mounted on the Port of LA EO-1 satellite. The hyperspectral image is 120 × 120 in size and consists of 242 bands, and after removing uncalibrated and noisy bands, the remaining 145 bands were used for the experiment. ALI also provides PAN images of the same scene with a spatial resolution of 30 m 360 × 360 in size. In our experiments, we resized the LRHS image to 90 × 90 × 145 by interpolation.

C. Qualitative Assessment Metrics

1) SAM: SAM is an algorithm used to evaluate the SSIM between two images, focusing mainly on the structural information of the images. It is formulated as follows:

$$\mathrm{SAM}(\mathbf{X},\mathbf{F}) = \frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W}\arccos\left(\frac{\mathbf{X}(:,i,j)\cdot\mathbf{F}(:,i,j)}{\|\mathbf{X}(:,i,j)\|_2\|\mathbf{F}(:,i,j)\|_2}\right) \quad (22)$$

where $\mathbf{X}$ represents the reference image, and $\mathbf{X}(:,i,j)$ is the (i,j)th pixel of $\mathbf{X}$.

2) PSNR: PSNR calculates the mse [see (17)] between the original image and the reconstructed image, which is usually expressed in decibels (dB). The higher the PSNR value, the lower the distortion, i.e., the smaller the difference between the reconstructed image and the original image, and the better the image quality

$$\mathrm{PSNR}(\mathbf{X},\mathbf{F}) = 10\log_{10}\frac{\mathrm{MAX}^2(\mathbf{F})}{\mathrm{mse}(\mathbf{X},\mathbf{F})}. \quad (23)$$

3) RMSE: RMSE calculates the difference between each pixel between the reconstructed and original images

$$\mathrm{RMSE}(\mathbf{X},\mathbf{F}) = \sqrt{\frac{1}{BHW}\sum_{t=1}^{B}\sum_{i=1}^{H}\sum_{j=1}^{W}(\mathbf{X}(t,i,j)-\mathbf{F}(t,i,j))^2}. \quad (24)$$

4) ERGAS: ERGAS is a comprehensive evaluation metric that calculates the RMSE and the average spatial correlation (MEAN) for each band. The former computes the difference between the reconstructed image and the original image, and the latter computes the loss of spatial correlation in the reconstructed image

$$\mathrm{ERGAS}(\mathbf{X},\mathbf{F}) = \frac{100}{r}\sqrt{\frac{1}{B}\sum_{t=1}^{B}\frac{\mathrm{mse}(\mathbf{X}(t,:,:),\mathbf{F}(t,:,:))}{\mathrm{MEAN}^2(\mathbf{F}(t,:,:))}} \quad (25)$$

where $\mathbf{X}(t,:,:)$ is the tth band of $\mathbf{X}$, and r is the sampling factor.

5) SSIM: SSIM measures the average SSIM in the spatial domain between the fused and reference images

$$\mathrm{SSIM}(\mathbf{X},\mathbf{F}) = \frac{1}{B}\sum_{t=1}^{B}\mathrm{SSIM}(\mathbf{X}(t,:,:),\mathbf{F}(t,:,:)) \quad (26)$$

$$\mathrm{SSIM}(x,y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2+\mu_y^2+C_1)(\sigma_x^2+\sigma_y^2+C_2)} \quad (27)$$

where x and y are the two images, μ_x and μ_y are the means of x and y, σ_x and σ_x are the standard deviations of x and y, σ_{xy} is the covariance between x and y, and C_1 and C_2 are constants for numerical stability.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:40:21 UTC from IEEE Xplore. Restrictions apply.

6) D_s: D_s is a spatial distortion index calculated as

$$D_s = \sqrt[q]{\frac{1}{B}\sum_{t=1}^{B}\left|\mathrm{UIQI}(\mathbf{F}(t,:,:),\mathbf{P}) - \mathrm{UIQI}\left(\mathbf{Y}(t,:,:),\widetilde{\mathbf{P}}\right)\right|^q} \tag{28}$$

where q is set to 1 and $\widetilde{\mathbf{P}}$ represents the spatial degradation result of the PAN image, which has the same spatial resolution as the LRHS image. The *universal image quality index* (UIQI) [73] is calculated as the dissimilarity between LRHS and generated HRHS images.

7) D_λ: The spectral distortion index D_λ is computed as the spectral difference between the reconstructed and reference images

$$D_\lambda = \sqrt[p]{\frac{1}{B(B-1)}\sum_{t_1=1}^{B}\sum_{t_2=1,t_2\neq t_1}^{B}\left|Q_1^{t_1,t_2} - Q_2^{t_1,t_2}\right|^p} \tag{29}$$

where p is set to 1, $Q_1^{t_1,t_2} = \mathrm{UIQI}(\mathbf{Y}(t_1,:,:),\mathbf{Y}(t_2,:,:))$ and $Q_2^{t_1,t_2} = \mathrm{UIQI}(\mathbf{F}(t_1.:,:),\mathbf{F}(t_2,:,:))$.

8) QNR: The result of QNR is related to D_s and D_λ which measures the overall quality of the generated image. Its expectation value is 1, meaning that the generated image has neither spatial loss nor spectral loss

$$\mathrm{QNR} = (1-D_\lambda)^\alpha(1-D_s)^\beta \tag{30}$$

where α and β are two coefficients.

D. Network Training and Experimental Setup

MFT-GAN is constructed based on the transformer, which requires a high amount of data for training, so we adopt the following approach to produce training and testing samples. For the simulated dataset, we produce LRHS images by Gaussian blur downsampling (Gaussian filter with a mean of 0, a standard deviation of 2, and a sampling rate r of 4) to the original images and create PAN images through spectral downsampling according to the Wald protocol. We use a convolutional-like operation to produce the training samples, extracting overlapping patches of 8×8 and 32×32 from the LRHS and PAN images, respectively. The step size of the extracted patches is 16 for the CAVE dataset, 14 for the Chikusei dataset, and 3 for the Bostwana dataset. For the real dataset, overlapping patches are 12×12 and 48×48 from the LRHS and PAN images, respectively. The patches are taken with a step size of 4. The original LRHS and PAN images are used as test images.

MFT-GAN is implemented by the PyTorch framework and trained in an iterative alternating manner on a single GeForce GTX 3060 12-GB graphics processor. The parameters of the spatial and spectral discriminators are updated at each iteration of the generator which yields a trained model after 300 iterations of training. Here, AdamW is used as an optimizer with a learning rate of 0.0001 and a batch size of 16. The tradeoff parameters in our loss function (15) are set to be $\lambda_1 = 1$, $\lambda_2 = 1$, and $\lambda_3 = 0.001$. The other parameters have default values in the AdamW optimizer.

TABLE II

ABLATION EXPERIMENTS ON HYBRID LOSS FUNCTIONS ON THE CAVE DATASET. "✓" REPRESENTS THAT THE LOSS TERM IS USED

Loss Items			Qualitative Assessment Metrics				
$L_{spectral}$	$L_{spatial}$	L_{adv}	SAM	PSNR	RMSE	ERGAS	SSIM
✓	✓	✓	**4.508**	**40.410**	**0.010**	1.254	**0.977**
	✓	✓	26.075	30.144	0.031	4.261	0.788
✓		✓	4.516	31.731	0.028	3.421	0.910
✓	✓		4.527	40.246	**0.010**	**1.236**	**0.977**
✓			4.511	31.730	0.028	3.418	0.910
	✓		31.202	28.247	0.038	5.262	0.711
		✓	51.890	20.010	0.102	58.813	0.403
Best value			0	$+\infty$	0	0	1

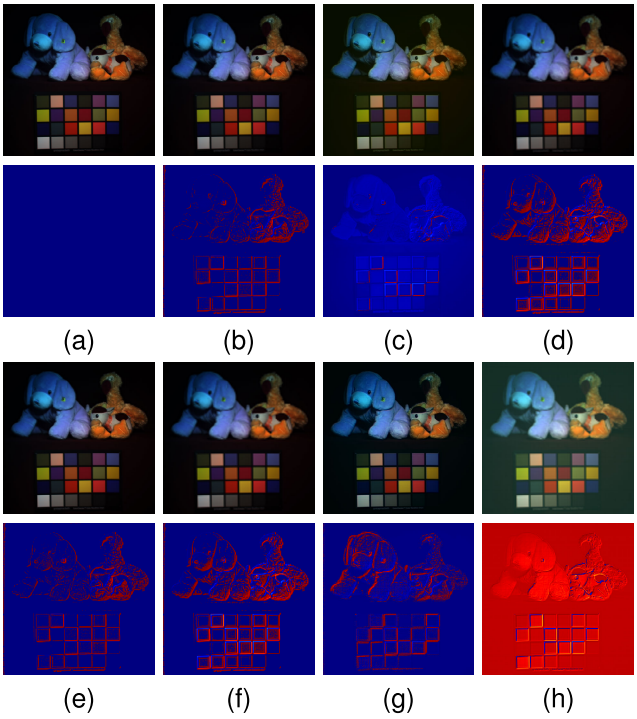


Fig. 6. Visualization of ablation experiments on the CAVE dataset using MTF-GAN. (Top row) Reconstructed images. (Bottom row) Error images. To facilitate the representation, we have used "1" to indicate that a loss term was used and "0" to indicate that a loss term was not applicable in the visualization results and have coded them in the order of L_{spectral}, L_{spatial}, and L_{adv}. For example, "011" indicates that the spectral loss term (L_{spectral}) was not used, while the spatial and adversarial loss terms (L_{spatial} and L_{adv}) were used. (a) Ground truth. (b) MFT-GAN. (c) 011. (d) 101. (e) 110. (f) 100. (g) 010. (h) 001. The three channels selected for the false color image are red: 29, green: 19, and blue: 9.

E. Ablation Studies

1) Loss Items: In (15), we present a hybrid loss function with three loss terms to train our MFT-GAN. To validate the impact and performance of each loss term, we performed six experiments in which all the loss terms, except for the loss function, were trained and tested on the same framework and dataset. Table II and Fig. 6 illustrate the above ablation experiments and visualization results performed on the CAVE dataset. In the absence of spectral loss (L_{spectral}) and spatial loss (L_{spatial}), all the metrics sharply worsen. This

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:40:21 UTC from IEEE Xplore. Restrictions apply.

TABLE III
ABLATION EXPERIMENT ON MFT-GAN STRUCTURE ON THE CAVE DATASET

Network				Qualitative Assessment Metrics				
Generator	Spectral Discriminator	Spatial Discriminator	Total Parameters / FLOPs	SAM	PSNR	RMSE	ERGAS	SSIM
ResBlock	Spectral Transformer	Spatial Transformer	3.642M / 2.306G	7.167	33.188	0.022	2.509	0.918
CBAM	Spectral Transformer	Spatial Transformer	0.917M / 1.956G	4.563	39.247	0.011	1.437	0.970
Multihead Cross Attention	CAB	SAB	1.631M / 1.116G	4.513	40.327	**0.010**	1.271	**0.977**
ResBlock	CAB	SAB	3.090M / 2.283G	5.909	35.185	0.017	2.152	0.910
CBAM	CAB	SAB	1.404M / 0.894G	4.515	39.503	0.011	1.400	0.972
Multihead Cross Attention	Spectral Transformer	Spatial Transformer	2.183M / 1.139G	**4.508**	**40.410**	**0.010**	**1.254**	**0.977**
Best value				0	$+\infty$	0	0	1

is because our unsupervised model is heavily dependent on them for training. From the reconstruction and error images in Fig.6(c), (d), and (h), the generated images without these two losses suffer from significant spectral distortion and spatial information loss. When we remove the adversarial loss (L_{adv}), the fusion results slightly decrease in individual metrics and are confirmed by the visual results in Fig. 6(b) and (e). This proves that the adversarial learning of MFT-GAN is effective and improves the performance. When only a single loss is available, the overall result is unsatisfactory. And the reconstructed and error images of Fig. 6(f)–(h) are far from the ground truth. This is because equilibrium is achieved from a single loss function. Through comparison, it is evident that our proposed hybrid loss is effective and achieves better results.

2) Generator and Discriminator Networks: As shown in Fig. 2, MFT-GAN is a transformer-based network whose effectiveness is verified by modifications to MFT-GAN. Table III lists the quantitative results of the networks constructed by replacing the generator and discriminator with different modules on the CAVE dataset and the parameters and computational costs (FLOPs) of these networks. Specifically, we replace the multihead cross-attention in the generator with ResBlock [74] and the convolutional block attention module (CBAM) [75] and replace the spectral and spatial transformers in the discriminator with a channel attention block (CAB) [75] and a spatial attention block (SAB) [75], respectively. When the generator network is modified, the quantitative results of using CBAM outperform those of using ResBlock, and the results of using multihead cross-attention are even better. Moreover, the number of parameters and computational cost of using ResBlock are greater than those of using CBAM and multihead cross attention. ResBlock performs poorly in the feature interaction task, and it is evident that the attention mechanism provides improved performance at a lower computational cost than does pure convolution. The reason why multihead cross-attention provides better results than CABM is that multihead cross-attention can capture the global features of an image through cross-computation. When the discriminator network is modified and the generator is kept constant, both the number of parameters and the computational cost of applying CAB and SAB to the discriminator are lower than those of MFT-GAN, but their quantitative results are slightly lower than those of MFT-GAN. This proves that applying the attention mechanism in the discriminator is effective. When both the generator and the discriminator are modified, the results of CBAM are clearly better than those of ResBlock, and its computational cost is lower than that of ResBlock. Our proposed method yields slightly better quantitative results. From the above comparison, it can be concluded that the MFT-GAN framework is effective and that the combination of an organic GAN and a transformer indeed improves the performance.

TABLE IV
QUALITY ASSESSMENT RESULTS OF IMAGES GENERATED BY MFT-GAN ON THE CAVE DATASET

Method	Qualitative Assessment Metrics				
	SAM	PSNR	RMSE	ERGAS	SSIM
GSA	5.964	37.336	0.014	3.652	0.956
HySure	7.162	34.906	0.015	4.887	0.946
FUSE	5.540	34.670	0.012	4.909	0.953
CNMF	6.799	37.006	0.013	4.082	0.964
GDD	5.207	31.886	0.017	6.737	0.920
Pan-GAN	6.507	33.242	0.029	2.542	0.928
UCGAN	7.547	32.854	0.024	5.815	0.924
IU2PNet	4.500	31.733	0.028	3.431	0.910
PLRDiff	5.622	40.022	**0.010**	2.924	**0.977**
MFT-GAN	**4.508**	**40.410**	**0.010**	**1.254**	**0.977**

TABLE V
QUALITY ASSESSMENT RESULTS OF IMAGES GENERATED BY MFT-GAN ON THE CHIKUSEI DATASET

Method	Qualitative Assessment Metrics				
	SAM	PSNR	RMSE	ERGAS	SSIM
GSA	4.619	40.115	0.010	6.398	0.957
HySure	6.228	36.464	0.015	9.539	0.904
FUSE	4.683	38.286	0.012	6.770	0.944
CNMF	4.714	37.969	0.012	6.258	0.928
GDD	4.935	35.142	0.017	7.843	0.890
Pan-GAN	6.709	35.189	0.017	4.172	0.857
UCGAN	5.799	37.433	0.013	8.218	0.911
IU2PNet	4.221	36.023	0.015	3.389	0.897
PLRDiff	5.215	40.235	0.009	6.607	0.946
MFT-GAN	**4.237**	**41.785**	**0.008**	**2.759**	**0.964**

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:40:21 UTC from IEEE Xplore. Restrictions apply.

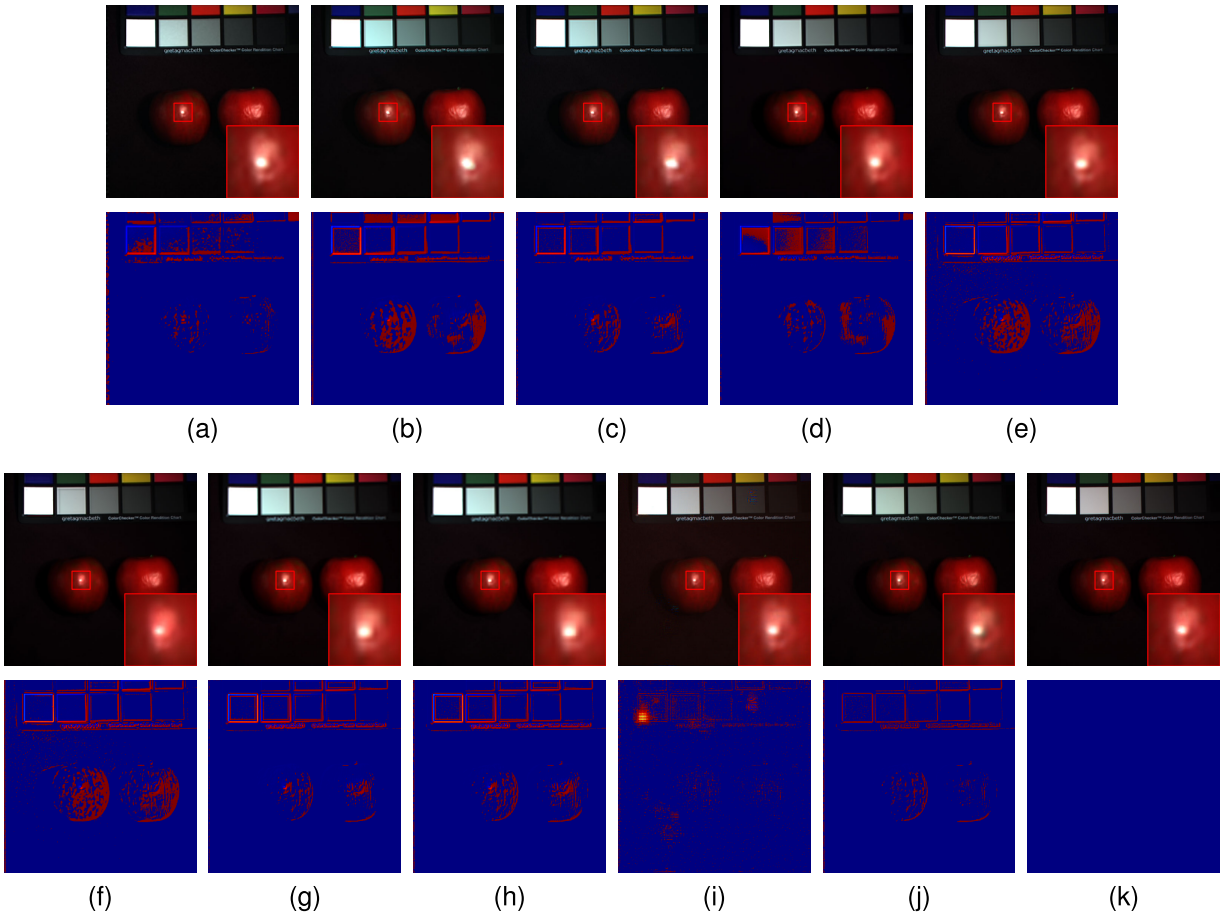


Fig. 7. Visualization of the CAVE dataset from different methods. (Top row) Reconstructed images. (Bottom row) Error images. (a) GSA. (b) HySure. (c) FUSE. (d) CNMF. (e) GDD. (f) Pan-GAN. (g) UCGAN. (h) IU2PNet. (i) PLRDiff. (j) MFT-GAN. (k) Ground truth. The three channels selected for the false color image are red: 29, green: 19, and blue: 9.

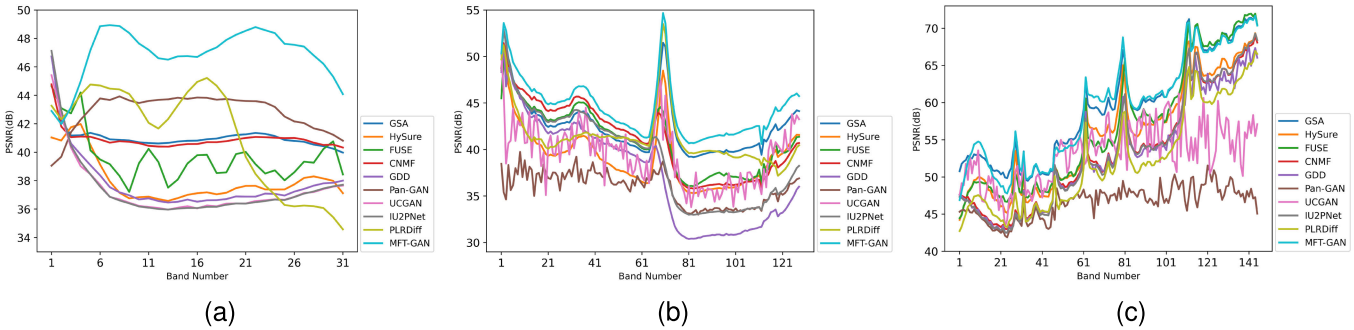


Fig. 8. PSNR is expressed as a function of the spectral band. (a) CAVE dataset. (b) Chikusei dataset. (c) Botswana dataset.

F. Experimental Results for the Simulated Datasets

To comprehensively evaluate the performance of MFT-GAN, we compared it with nine approaches, namely, the classical approaches GSA [14], A Convex Formulation for Hyperspectral Image Superresolution via Subspace-Based Regularization (HySure) [20], FUSE [38], and CNMF [36] and the deep-learning-based unsupervised approaches GDD [53], Pan-GAN [57], UCGAN [59], An interpretable unsupervised unrolling network for hyperspectral pansharpening (IU2PNet) [76], and PLRDiff [65]. These comparison approaches are consistent with their corresponding papers and source code. The results for the three simulated datasets are qualitatively and visually presented below.

1) CAVE: Table IV shows the mean results for the last 12 test images across the five quality evaluation metrics, and the best results are marked in bold. Our proposed method clearly outperforms the other methods in terms of all the metrics. This indicates that the MFT-GAN method can complete hyperspectral pansharpening with minimal distortion. Furthermore, both the PLRDiff method and our proposed method outperform the remaining four deep-learning-based unsupervised methods. In comparison to the PLRDiff method, our method also has higher spectral fidelity. This further validates the superiority of the proposed framework.

The visualization results of a test image from the CVAE dataset ("real_and_fake_apples") are shown in Fig. 7 of all the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:40:21 UTC from IEEE Xplore. Restrictions apply.

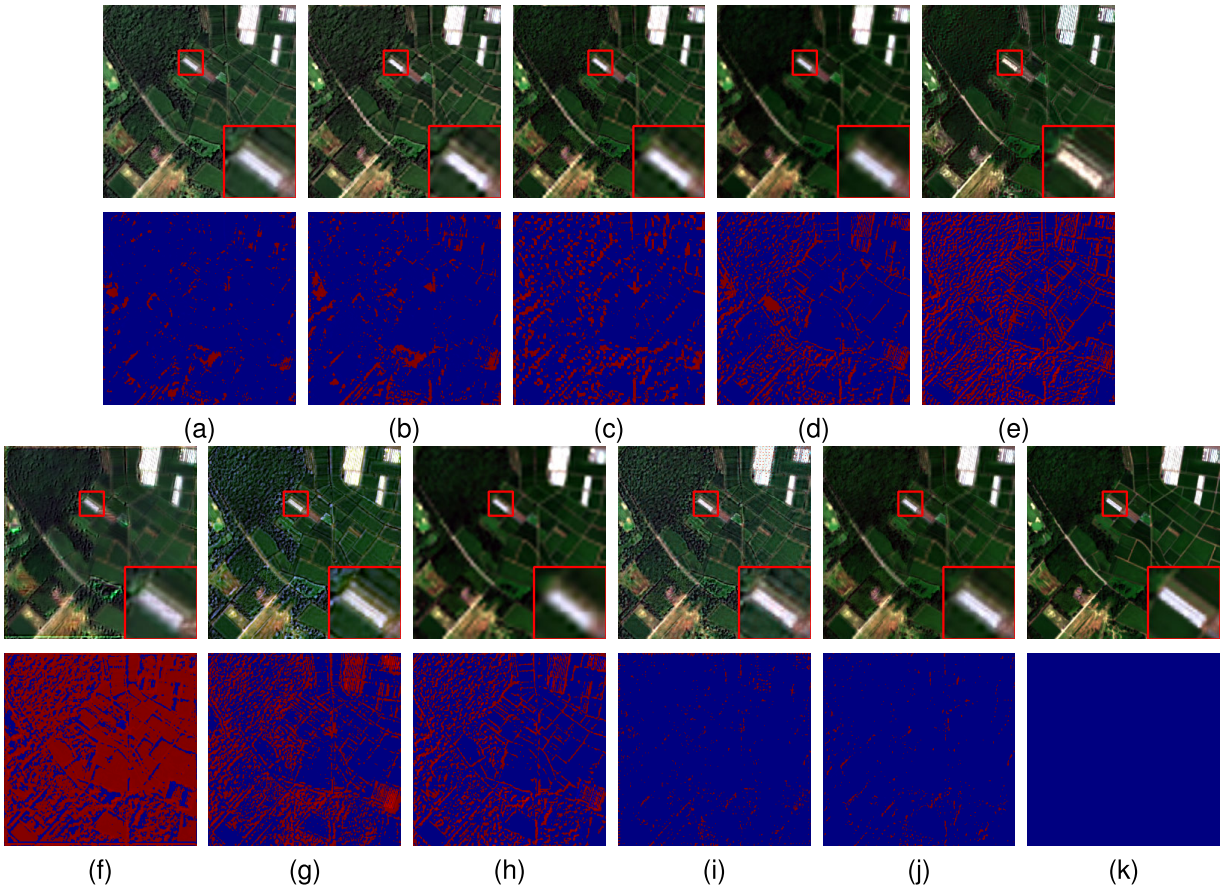


Fig. 9. Visualization of the Chikusei dataset from different methods. (Top row) Reconstructed images. (Bottom row) Error images. (a) GSA. (b) HySure. (c) FUSE. (d) CNMF. (e) GDD. (f) Pan-GAN. (g) UCGAN. (h) IU2PNet. (i) PLRDiff. (j) MFT-GAN. (k) Ground truth. The three channels selected for the false color image are red: 60, green: 40, and blue: 20.

methods. The first row shows the false color images for the reconstructed images. The second row shows the error images. In terms of reconstructed images, the CNMF, Pan-GAN, and UCGAN methods produce images that lose detailed information to varying degrees. The reconstructed images from the other methods are more similar to the reference image. From the error images, it is observed that the HySure, CNMF, GDD, and Pan-GAN methods exhibit the highest error rates, while the FUSE, UCGAN, and IU2PNet methods demonstrate lower error rates. Conversely, the GSA, PLRDiff, and MFT-GAN methods produce the least errors. In addition, we present a comparison of the quality of the fused images across different spectral bands in Fig. 8(a). The PSNR values of all the traditional methods, as well as those of the GDD, UCGAN, and IU2PNet methods, are lower than those of the Pan-GAN, PLRDiff, and MFT-GAN methods. This suggests that the former are more susceptible to distortion in bands 5–20. Overall, the PSNR curve of MFT-GAN dominates, further affirming the effectiveness of MFT-GAN on the CAVE dataset.

2) Chikusei: Table V displays the average quantization results for the final 19 test images in the Chikusei dataset. For the SAM and ERGAS metrics, the IU2PNet and MFT-GAN methods exhibit the best performance. Conversely, for the remaining metrics, the PLRDiff and MFT-GAN methods yield superior results. Overall, MFT-GAN consistently outperforms other methods across all the metrics on the Chikusei dataset, thus demonstrating its effectiveness.

TABLE VI
QUALITY ASSESSMENT RESULTS OF IMAGES GENERATED BY MFT-GAN ON THE BOTSWANA DATASET

Method	Qualitative Assessment Metrics				
	SAM	PSNR	RMSE	ERGAS	SSIM
GSA	2.574	47.293	0.004	3.203	0.984
HySure	3.142	49.188	0.003	2.643	0.857
FUSE	2.609	44.694	0.005	3.825	0.976
CNMF	3.760	42.922	0.007	5.248	0.991
GDD	2.637	47.036	0.004	3.531	0.983
Pan-GAN	4.418	44.955	0.005	4.720	0.819
UCGAN	3.147	49.544	0.003	7.957	0.989
IU2PNet	**2.522**	46.886	0.004	1.635	0.982
PLRDiff	3.868	47.121	0.004	4.360	0.982
MFT-GAN	**2.522**	**51.649**	**0.002**	**0.990**	**0.994**

Fig. 9 displays the reconstructed and error images for all the methods on the Chikusei dataset. From the reconstructed images, it is notable that the Pan-GAN and IU2PNet methods produce blurrier images. The CNMF, UCGAN, and PLRDiff methods suffer from the loss of detailed information while the remaining methods exhibit images closer to the reference image. Analyzing the error images, there are obvious

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:40:21 UTC from IEEE Xplore. Restrictions apply.

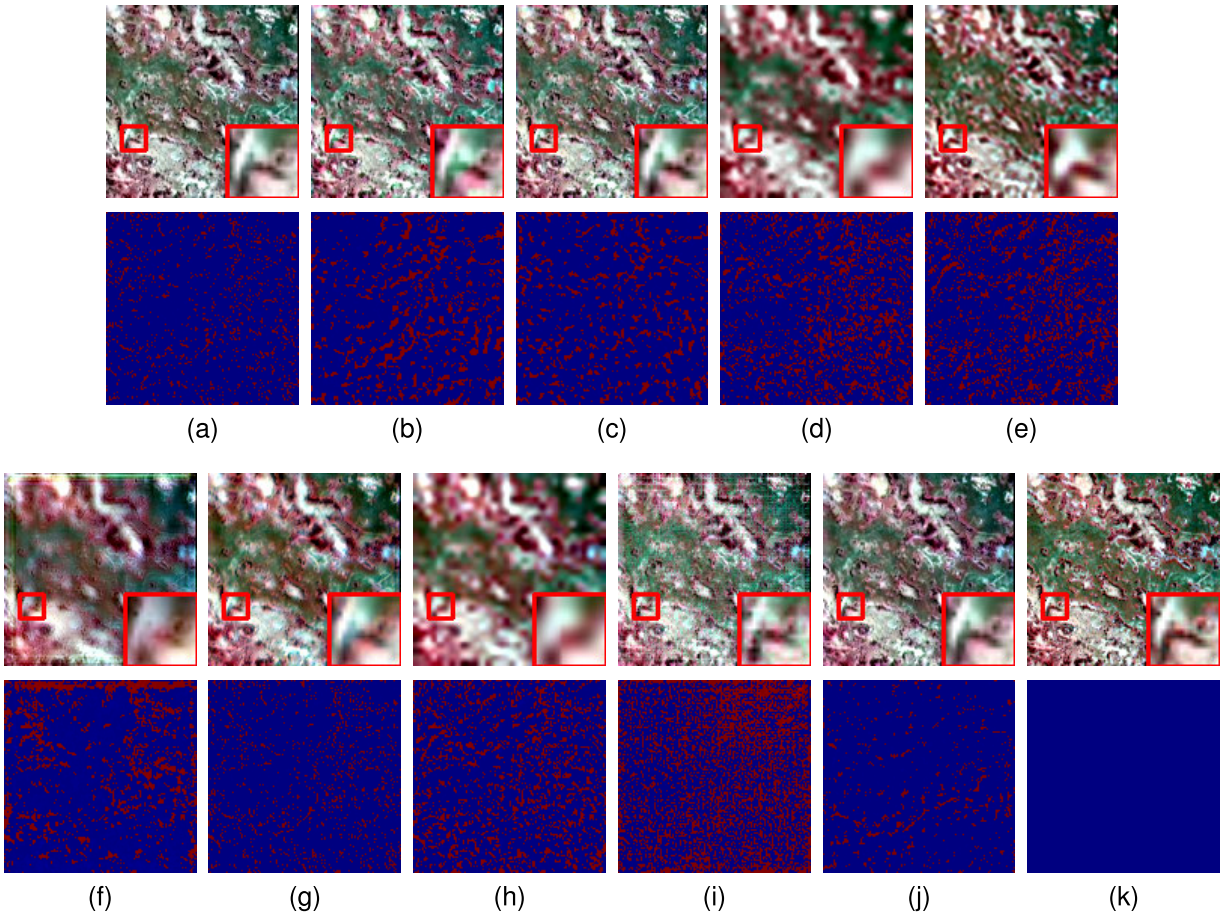


Fig. 10. Visualization of the Botswana dataset from different methods. (Top row) Reconstructed images. (Bottom row) Error images. (a) GSA. (b) HySure. (c) FUSE. (d) CNMF. (e) GDD. (f) Pan-GAN. (g) UCGAN. (h) IU2PNet. (i) PLRDiff. (j) MFT-GAN. (k) Ground truth. The three channels selected for the false color image are red: 29, green: 20, and blue: 12.

TABLE VII
QUALITY ASSESSMENT RESULTS OF IMAGES GENERATED BY MFT-GAN ON THE LA DATASET

Method	D_s	D_λ	QNR
Best Values	0	0	1
GSA	0.049	0.072	0.881
HySure	0.031	0.054	0.915
FUSE	0.033	0.044	0.923
CNMF	0.043	0.046	0.911
GDD	0.043	0.039	0.919
Pan-GAN	0.044	**0.032**	0.924
UCGAN	0.177	0.115	0.728
IU2PNet	0.069	0.060	0.874
PLRdiff	0.035	0.069	0.897
MFT-GAN	**0.027**	0.036	**0.936**

differences from the reference image in the FUSE, CNMF, GDD, Pan-GAN, UCGAN, and IU2PNet methods, whereas the PLRDiff and MFT-GAN methods yield results closer to the reference image. Fig. 8(b) shows the PSNR curves for all the methods on the Chikusei dataset. Notably, all the methods perform well in all the bands, with MFT-GAN achieving the highest PSNR results. In conclusion, MFT-GAN yields good results in terms of both quantization and visualization.

3) Botswana: Table VI shows the average results for the final 12 test images in the Botswana dataset. For the SAM and ERGAS metrics, the GSA, IU2PNet, and MFT-GAN methods outperform the others, with the IU2PNet method showing the best results. In terms of PSNR and RMSE, the HySure, UCCGAN, and MFT-GAN methods perform better than others. In addition, the CNMF and MFT-GAN methods demonstrate superior results in SSIM. Overall, MFT-GAN achieves the best results across most metrics on the Botswana dataset, proving its effectiveness.

The visualization results for all the methods applied to the Botswana dataset are presented in Fig. 10. By observing the reconstructed images, it becomes evident that CNMF, GDD, Pan-GAN, UCGAN, and IU2PNet exhibit blurriness and varying degrees of detail information loss, while the remaining methods closely approximate the reference image. When looking at the error images, it is clear that GSA, UCGAN, and MFT-GAN are very close to the reference image, while the rest of the methods give slightly worse results compared with the reference image. Fig. 8(c) shows the PSNR results of all the methods for all the bands on the Botswana dataset. All the methods achieve satisfactory results in each band. Specifically, in bands 1–81, GSA and MFT-GAN offer the best results. In bands 81–145, the curves of GSA and MFT-GAN largely overlap and exhibit an advantage compared

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:40:21 UTC from IEEE Xplore. Restrictions apply.

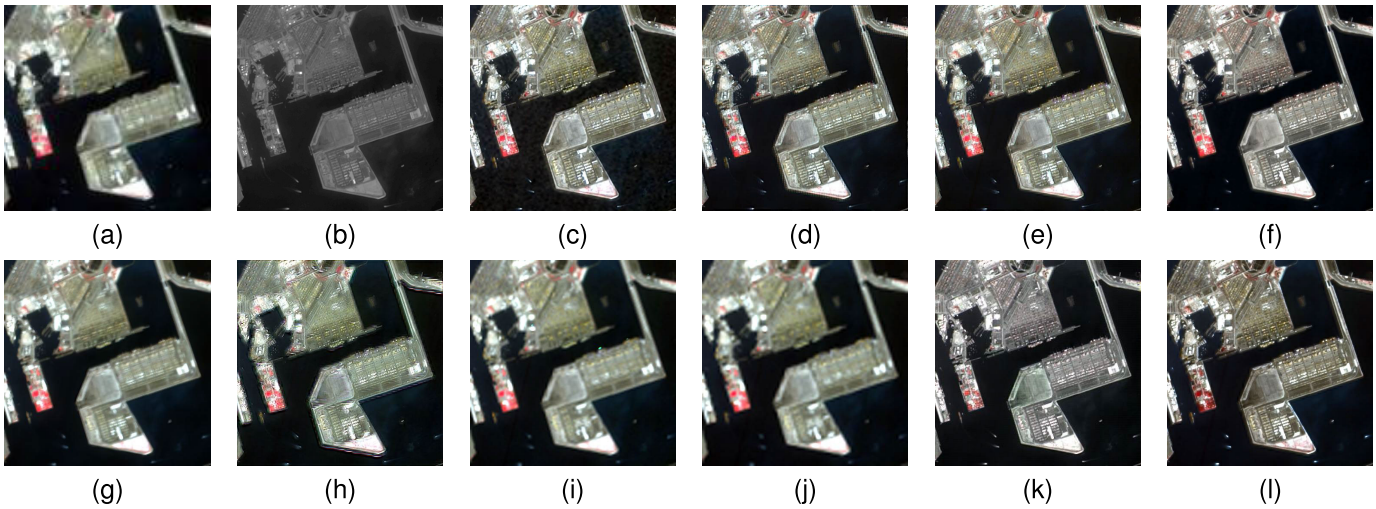


Fig. 11. Visualization of the LA dataset from different methods. (a) Real LRHS image. (b) Real PAN image. (c) GSA. (d) HySure. (e) FUSE. (f) CNMF. (g) GDD. (h) Pan-GAN. (i) UCGAN. (j) IU2PNet. (k) PLRDiff. (l) MFT-GAN. The three channels selected for the false color image are red: 29, green: 20, and blue: 12.

with other methods. Generally, MFT-GAN yields better results on the Botswana dataset, demonstrating its performance and effectiveness.

G. Experimental Results for the Real Dataset

Table VII lists the results of all the competing methods on the LA dataset. Due to the complex feature information in the LA dataset, pansharpening methods struggle to capture the full feature details. The experimental results clearly show that FUSE, Pan-GAN, and MFT-GAN exhibit relatively strong performance. Pan-GAN achieves the best results for the spectral distortion metric D_λ, while MFT-GAN outperforms the others for the remaining metrics. This demonstrates the effectiveness and superior performance of the proposed method compared with others on the real dataset. The visualization results for the LA dataset are illustrated in Fig. 11. The GDD, UCGAN, and IU2PNet methods produce relatively blurred images and suffer from a loss of detailed information. In conclusion, the results of MFT-GAN show its advantages in terms of both quantitative metrics and visualization results, confirming the strength and effectiveness of the transformer module.

H. Computational Efficiency

All the methods were evaluated for their number of parameters, computational cost, and inference time (PARAMS, FLOPs, TIME) on the CAVE dataset, using an LRHS image of size $16 \times 16 \times 31$ and a PAN image of size $64 \times 64 \times 1$. All the traditional methods were implemented on the CPU using MATLAB. Deep-learning-based methods are executed on a GPU. The testing computer is equipped with an AMD Ryzen 7 5800H processor running at 3.2 GHz and 16 GB of RAM. The efficiency of the different methods is shown in Table VIII. Deep-learning-based methods usually require more time than traditional methods. The GDD method does not require many samples to train and gets the fusion result in one training, hence it takes a longer time for inference. The PLRdiff method also has a longer inference time because the inverse process of the diffusion model involves a substantial number of computations. Conversely, the remaining deep learning methods generally have shorter inference times. In terms of PARAMS and FLOPs, the PLRdiff and IU2PNet methods have larger numbers of parameters and computational costs, while GDD and UCGAN have smaller ones. Overall, the efficiency of the MFT-GAN method has some advantages over the more complex structure of the CNN-based IU2PNet method and the diffusion-model-based PLRdiff method.

TABLE VIII
PARAMETER COUNTS, FLOPS, AND TESTING TIMES OF THE COMPARED METHODS ON THE CAVE DATASETS

Method	PARAMS	FLOPs	TIME
GSA	-	-	0.219s
HySure	-	-	2.635s
FUSE	-	-	0.454s
CNMF	-	-	0.515s
GDD	480.449K	797.139K	568.5s
Pan-GAN	1.679M	6.880G	0.088s
UCGAN	266.253K	1.092G	0.036s
IU2PNet	8.409M	18.840G	2.850s
PLRdiff	390.952M	44.777G	40.929s
MFT-GAN	1.535M	4.663G	0.681s

V. CONCLUSION

In this work, we introduce an unsupervised hyperspectral pansharpening network based on multiscale feature guidance, called MFT-GAN. Taking advantage of the transformer, this approach achieves feature interaction at the same scale and feature fusion at different scales and improves the preservation of spectral and spatial information through adversarial learning. Specifically, MFT-GAN consists of a generator and two discriminators (a spectral transformer discriminator and a spatial transformer discriminator). The generator network includes two branches: the MFGB and FIFB branches. MFGB

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:40:21 UTC from IEEE Xplore. Restrictions apply.

uses CNNs to acquire spectral information at different scales and spatial information from LRHSs and PAN images using level-by-level up- and downsampling, respectively. FIFB integrates the spectral and spatial information of the same scale in the former by the designed multihead cross-attention and then fuses the interaction results of different scales by level-by-level upsampling to obtain the final fused image. Spectral transformer and spatial transformer discriminators are used to form an adversarial game state with the generator so that spectral and spatial information preservation can be improved. Finally, the adversarial training of the MFT-GAN is combined with a hybrid loss function whose experimental results confirm its effectiveness.

ACKNOWLEDGMENT

The authors would like to express their gratitude to all the organizations and researchers who generously provided the experimental data used in this study. They also extend their thanks to the authors who kindly provided the source codes of the comparison methods discussed in this article. In addition, they appreciate the valuable comments and suggestions provided by the editors and reviewers, which greatly contributed to the improvement of this work.

REFERENCES

[1] L. Loncan et al., "Hyperspectral pansharpening: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 3, no. 3, pp. 27–46, Sep. 2015.

[2] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[3] M. Fauvel, Y. Tarabalka, J. A. Benediktsson, J. Chanussot, and J. C. Tilton, "Advances in spectral–spatial classification of hyperspectral images," *Proc. IEEE*, vol. 101, no. 3, pp. 652–675, Mar. 2013.

[4] J. Liu, Z. Wu, J. Li, A. Plaza, and Y. Yuan, "Probabilistic-kernel collaborative representation for spatial–spectral hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 2371–2384, Apr. 2016.

[5] J. M. Bioucas-Dias et al., "Hyperspectral unmixing overview: Geometrical, statistical, and sparse regression-based approaches," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 354–379, Apr. 2012.

[6] R. Heylen, M. Parente, and P. Gader, "A review of nonlinear hyperspectral unmixing methods," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 1844–1868, Jun. 2014.

[7] L. Gao, Z. Wang, L. Zhuang, H. Yu, B. Zhang, and J. Chanussot, "Using low-rank representation of abundance maps and nonnegative tensor factorization for hyperspectral nonlinear unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2022, Art. no. 5504017, doi: 10.1109/TGRS.2021.3065990.

[8] L. Gao, Z. Han, D. Hong, B. Zhang, and J. Chanussot, "CyCU-Net: Cycle-consistency unmixing network by learning cascaded autoencoders," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022, Art. no. 5503914, doi: 10.1109/TGRS.2021.3064958.

[9] G. Cheng, P. Zhou, and J. Han, "Learning rotation-invariant convolutional neural networks for object detection in VHR optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7405–7415, Dec. 2016.

[10] L. Gao, D. Wang, L. Zhuang, X. Sun, M. Huang, and A. Plaza, "BS^{3}LNet: A new blind-spot self-supervised learning network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–18, 2023, Art. no. 5504218, doi: 10.1109/TGRS.2023.3246565.

[11] L. Gao, X. Sun, X. Sun, L. Zhuang, Q. Du, and B. Zhang, "Hyperspectral anomaly detection based on chessboard topology," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–16, 2023, Art. no. 5505016, doi: 10.1109/TGRS.2023.3249748.

[12] V. P. Shah, N. H. Younan, and R. L. King, "An efficient pan-sharpening method via a combined adaptive PCA approach and contourlets," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 5, pp. 1323–1335, May 2008.

[13] T.-M. Tu, P. S. Huang, C.-L. Hung, and C.-P. Chang, "A fast intensity–hue–saturation fusion technique with spectral adjustment for IKONOS imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 1, no. 4, pp. 309–312, Oct. 2004.

[14] B. Aiazzi, S. Baronti, and M. Selva, "Improving component substitution pansharpening through multivariate regression of MS+Pan data," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 10, pp. 3230–3239, Oct. 2007.

[15] J. G. Liu, "Smoothing filter-based intensity modulation: A spectral preserve image fusion technique for improving spatial details," *Int. J. Remote Sens.*, vol. 21, no. 18, pp. 3461–3472, Jan. 2000.

[16] J. Zhou, D. L. Civco, and J. A. Silander, "A wavelet transform method to merge Landsat TM and SPOT panchromatic data," *Int. J. Remote Sens.*, vol. 19, no. 4, pp. 743–757, Jan. 1998.

[17] H. Shen, X. Meng, and L. Zhang, "An integrated framework for the spatio–temporal–spectral fusion of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7135–7148, Dec. 2016.

[18] C. Ballester, V. Caselles, L. Igual, J. Verdera, and B. Rougé, "A variational model for P+XS image fusion," *Int. J. Comput. Vis.*, vol. 69, no. 1, pp. 43–58, Aug. 2006.

[19] F. Palsson, J. R. Sveinsson, and M. O. Ulfarsson, "A new pansharpening algorithm based on total variation," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 1, pp. 318–322, Jan. 2014.

[20] M. Simoes, J. Bioucas-Dias, L. B. Almeida, and J. Chanussot, "A convex formulation for hyperspectral image superresolution via subspace-based regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 3373–3388, Jun. 2015.

[21] A. Sellami and S. Tabbone, "Deep neural networks-based relevant latent representation learning for hyperspectral image classification," *Pattern Recognit.*, vol. 121, Jan. 2022, Art. no. 108224.

[22] X. Cao, J. Yao, Z. Xu, and D. Meng, "Hyperspectral image classification with convolutional neural network and active learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4604–4616, Jul. 2020.

[23] Y. Su, L. Gao, M. Jiang, A. Plaza, X. Sun, and B. Zhang, "NSCKL: Normalized spectral clustering with kernel-based learning for semisupervised hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 53, no. 10, pp. 6649–6662, Jul. 2023.

[24] M. H. Hesamian, W. Jia, X. He, and P. Kennedy, "Deep learning techniques for medical image segmentation: Achievements and challenges," *J. Digit. Imag.*, vol. 32, no. 4, pp. 582–596, Aug. 2019.

[25] W. Yang, X. Zhang, Y. Tian, W. Wang, J.-H. Xue, and Q. Liao, "Deep learning for single image super-resolution: A brief review," *IEEE Trans. Multimedia*, vol. 21, no. 12, pp. 3106–3121, Dec. 2019.

[26] J.-F. Hu, T.-Z. Huang, L.-J. Deng, T.-X. Jiang, G. Vivone, and J. Chanussot, "Hyperspectral image super-resolution via deep spatiospectral attention convolutional neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 12, pp. 7251–7265, Dec. 2022.

[27] W. Dong, C. Zhou, F. Wu, J. Wu, G. Shi, and X. Li, "Model-guided deep hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 30, pp. 5754–5768, 2021.

[28] W. Wang, L.-J. Deng, R. Ran, and G. Vivone, "A general paradigm with detail-preserving conditional invertible network for image fusion," *Int. J. Comput. Vis.*, vol. 132, no. 4, pp. 1029–1054, Apr. 2024.

[29] M. Ciotola, S. Vitale, A. Mazza, G. Poggi, and G. Scarpa, "Pansharpening by convolutional neural networks in the full resolution framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2022, Art. no. 5408717, doi: 10.1109/TGRS.2022.3163887.

[30] M. Ciotola, G. Poggi, and G. Scarpa, "Unsupervised deep learning-based pansharpening with jointly enhanced spectral and spatial fidelity," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–17, 2023, Art. no. 5405417, doi: 10.1109/TGRS.2023.3299356.

[31] C. A. Laben and B. V. Brower, "Process for enhancing the spatial resolution of multispectral imagery using pan-sharpening," U.S. Patent 6 011 875, Jan. 4, 2000.

[32] G. Vivone et al., "A critical comparison among pansharpening algorithms," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 5, pp. 2565–2586, May 2015.

[33] F. Nencini, A. Garzelli, S. Baronti, and L. Alparone, "Remote sensing image fusion using the curvelet transform," *Inf. Fusion*, vol. 8, no. 2, pp. 143–156, Apr. 2007.

[34] P. Chavez et al., "Comparison of three different methods to merge multiresolution and multispectral data: LANDSAT TM and SPOT panchromatic: ABSTRACT," *AAPG Bull.*, vol. 74, pp. 295–303, Jan. 1990.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:40:21 UTC from IEEE Xplore. Restrictions apply.

[35] J. Liu, Z. Wu, L. Xiao, J. Sun, and H. Yan, "A truncated matrix decomposition for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 29, pp. 8028–8042, 2020.

[36] N. Yokoya, T. Yairi, and A. Iwasaki, "Coupled nonnegative matrix factorization unmixing for hyperspectral and multispectral data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 528–537, Feb. 2012.

[37] Y. Xu, Z. Wu, J. Chanussot, and Z. Wei, "Hyperspectral computational imaging via collaborative Tucker3 tensor decomposition," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 1, pp. 98–111, Jan. 2021.

[38] Q. Wei, N. Dobigeon, and J.-Y. Tourneret, "Fast fusion of multi-band images based on solving a Sylvester equation," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 4109–4121, Nov. 2015.

[39] Q. Xie, M. Zhou, Q. Zhao, Z. Xu, and D. Meng, "MHF-Net: An interpretable deep network for multispectral and hyperspectral image fusion," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 3, pp. 1457–1473, Mar. 2022.

[40] S. Xu, J. Zhang, Z. Zhao, K. Sun, J. Liu, and C. Zhang, "Deep gradient projection networks for pan-sharpening," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 1366–1375.

[41] D. Shen, J. Liu, Z. Wu, J. Yang, and L. Xiao, "ADMM-HFNet: A matrix decomposition-based deep approach for hyperspectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2022, Art. no. 5513417, doi: 10.1109/TGRS.2021.3112181.

[42] X. Meng, N. Wang, F. Shao, and S. Li, "Vision transformer for pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2022, Art. no. 5409011, doi: 10.1109/TGRS.2022.3168465.

[43] M. Zhou, J. Huang, Y. Fang, X. Fu, and A. Liu, "Pan-Sharpening with customized transformer and invertible neural network," in *Proc. AAAI Conf. Artif. Intell.*, Jun. 2022, vol. 36, no. 3, pp. 3553–3561.

[44] W. G. C. Bandara and V. M. Patel, "HyperTransformer: A textural and spectral feature fusion transformer for pansharpening," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 1767–1777.

[45] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[46] Y. Long, X. Wang, M. Xu, S. Zhang, S. Jiang, and S. Jia, "Dual self-attention Swin transformer for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–12, 2023, Art. no. 5512012, doi: 10.1109/TGRS.2023.3275146.

[47] N. Liu, W. Li, X. Sun, R. Tao, and J. Chanussot, "Remote sensing image fusion with task-inspired multiscale nonlocal-attention network," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[48] G. Meng, J. Huang, Y. Wang, Z. Fu, X. Ding, and Y. Huang, "Progressive high-frequency reconstruction for pan-sharpening with implicit neural representation," in *Proc. AAAI Conf. Artif. Intell.*, 2024, vol. 38, no. 5, pp. 4189–4197.

[49] S. Jia, Z. Min, and X. Fu, "Multiscale spatial–spectral transformer network for hyperspectral and multispectral image fusion," *Inf. Fusion*, vol. 96, pp. 117–129, Aug. 2023.

[50] Q. Meng, W. Shi, S. Li, and L. Zhang, "PanDiff: A novel pansharpening method based on denoising diffusion probabilistic model," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–17, 2023, Art. no. 5611317, doi: 10.1109/TGRS.2023.3279864.

[51] Z. Cao, S. Cao, L.-J. Deng, X. Wu, J. Hou, and G. Vivone, "Diffusion model with disentangled modulations for sharpening multispectral and hyperspectral images," *Inf. Fusion*, vol. 104, Apr. 2024, Art. no. 102158.

[52] W. Huang, L. Xiao, Z. Wei, H. Liu, and S. Tang, "A new pan-sharpening method with deep neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 5, pp. 1037–1041, May 2015.

[53] T. Uezato, D. Hong, N. Yokoya, and W. He, "Guided deep decoder: Unsupervised image pair fusion," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Cham, Switzerland: Springer, 2020, pp. 87–102.

[54] Y. Qu, R. K. Baghbaderani, H. Qi, and C. Kwan, "Unsupervised pansharpening based on self-attention mechanism," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3192–3208, Apr. 2021.

[55] L. Gao, J. Li, K. Zheng, and X. Jia, "Enhanced autoencoders with attention-embedded degradation learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–17, 2023, Art. no. 5509417, doi: 10.1109/TGRS.2023.3267890.

[56] Q. Cao, L.-J. Deng, W. Wang, J. Hou, and G. Vivone, "Zero-shot semi-supervised learning for pansharpening," *Inf. Fusion*, vol. 101, Jan. 2024, Art. no. 102001.

[57] J. Ma, W. Yu, C. Chen, P. Liang, X. Guo, and J. Jiang, "Pan-GAN: An unsupervised pan-sharpening method for remote sensing image fusion," *Inf. Fusion*, vol. 62, pp. 110–120, Oct. 2020.

[58] H. Zhou, Q. Liu, and Y. Wang, "PGMAN: An unsupervised generative multiadversarial network for pansharpening," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 6316–6327, 2021.

[59] H. Zhou, Q. Liu, D. Weng, and Y. Wang, "Unsupervised cycle-consistent generative adversarial networks for pan sharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022, Art. no. 5408814, doi: 10.1109/TGRS.2022.3166528.

[60] X. Liu, X. Liu, H. Dai, X. Kang, A. Plaza, and W. Zu, "Mun-GAN: A multiscale unsupervised network for remote sensing image pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–18, 2023, Art. no. 5404018, doi: 10.1109/TGRS.2023.3288073.

[61] K. Zheng et al., "Coupled convolutional neural network with adaptive response function learning for unsupervised hyperspectral super resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2487–2502, Mar. 2021.

[62] J. Yao, D. Hong, J. Chanussot, D. Meng, X. Zhu, and Z. Xu, "Cross-attention in coupled unmixing nets for unsupervised hyperspectral super-resolution," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, Aug. 2020, pp. 208–224.

[63] J. Liu, Z. Wu, L. Xiao, and X.-J. Wu, "Model inspired autoencoder for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022, Art. no. 5522412, doi: 10.1109/TGRS.2022.3143156.

[64] J. Li, K. Zheng, Z. Li, L. Gao, and X. Jia, "X-shaped interactive autoencoders with cross-modality mutual learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5518317.

[65] X. Rui, X. Cao, L. Pang, Z. Zhu, Z. Yue, and D. Meng, "Unsupervised hyperspectral pansharpening via low-rank diffusion model," *Inf. Fusion*, vol. 107, Jul. 2024, Art. no. 102325.

[66] A. Vaswani et al., "Attention is all you need," *CoRR*, vol. abs/1706.03762, 2017. [Online]. Available: http://arxiv.org/abs/1706.0376

[67] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," *CoRR*, vol. abs/2010.11929, 2020. [Online]. Available: https://arxiv.org/abs/2010.11929

[68] I. Gulrajani, F. Ahmed, M. Arjovsky, V. Dumoulin, and A. C. Courville, "Improved training of Wasserstein GANs," *CoRR*, vol. abs/1704.00028, 2017. [Online]. Available: http://arxiv.org/abs/1704.00028

[69] A. Brock, J. Donahue, and K. Simonyan, "Large scale GAN training for high fidelity natural image synthesis," 2018, *arXiv:1809.11096*.

[70] T. Miyato, T. Kataoka, M. Koyama, and Y. Yoshida, "Spectral normalization for generative adversarial networks," 2018, *arXiv:1802.05957*.

[71] Z. Wang, E. P. Simoncelli, and A. C. Bovik, "Multiscale structural similarity for image quality assessment," in *Proc. 37th Asilomar Conf. Signals, Syst. Comput.*, vol. 2, Nov. 2003, pp. 1398–1402.

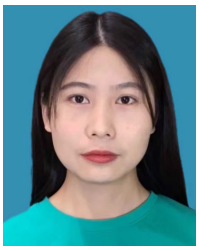
[72] L. Alparone, B. Aiazzi, S. Baronti, A. Garzelli, F. Nencini, and M. Selva, "Multispectral and panchromatic data fusion assessment without reference," *Photogramm. Eng. Remote Sens.*, vol. 74, no. 2, pp. 193–200, Feb. 2008.

[73] Z. Wang and A. C. Bovik, "A universal image quality index," *IEEE Signal Process. Lett.*, vol. 9, no. 3, pp. 81–84, Mar. 2002.

[74] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[75] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 3–19.

[76] J. Qu, W. Dong, Y. Li, S. Hou, and Q. Du, "An interpretable unsupervised unrolling network for hyperspectral pansharpening," *IEEE Trans. Cybern.*, vol. 53, no. 12, pp. 7943–7956, Jul. 2023.



Yanli Shang received the B.S. degree in network engineering from Anyang Institute of Technology, Anyang, China, in 2020, and the M.S. degree in computer technology from Jiangnan University, Wuxi, China, in 2023. She is currently pursuing the Ph.D. degree in electronic information with the School of Electronic Information, Wuhan University, Wuhan, China.

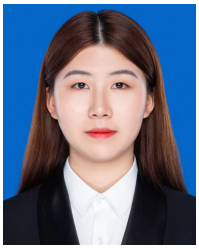
Her research interests include hyperspectral image fusion and deep learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:40:21 UTC from IEEE Xplore. Restrictions apply.



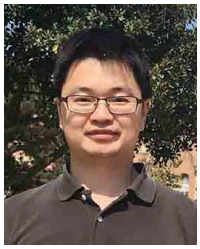
Jianjun Liu (Member, IEEE) received the B.S. degree in applied mathematics and the Ph.D. degree in pattern recognition and intelligence system from Nanjing University of Science and Technology, Nanjing, China, in 2009 and 2014, respectively.

From 2018 to 2020, he was a Post-Doctoral Researcher with the Department of Electrical Engineering, City University of Hong Kong, Hong Kong, China. He is currently an Associate Professor with the School of Artificial Intelligence and Computer Science, Jiangnan University, Wuxi, China. His research interests include hyperspectral image classification, super-resolution, spectral unmixing, sparse representation, computer vision, and pattern recognition.



Jingyi Zhang received the B.S. degree in computer science and technology from Henan Polytechnic University, Jiaozuo, China, in 2021. She is currently pursuing the master's degree in electronic information with the School of Artificial Intelligence and Computer Science, Jiangnan University, Wuxi, China.

Her research interests include deep learning and hyperspectral image fusion.



Zebin Wu (Senior Member, IEEE) received the B.S. and Ph.D. degrees in computer science from Nanjing University of Science and Technology, Nanjing, China, in 2003 and 2008, respectively.

He is currently a Professor with the School of Computer Science, Nanjing University of Science and Technology. His research interests include hyperspectral image processing, high-performance computing, and computer simulation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:40:21 UTC from IEEE Xplore. Restrictions apply.